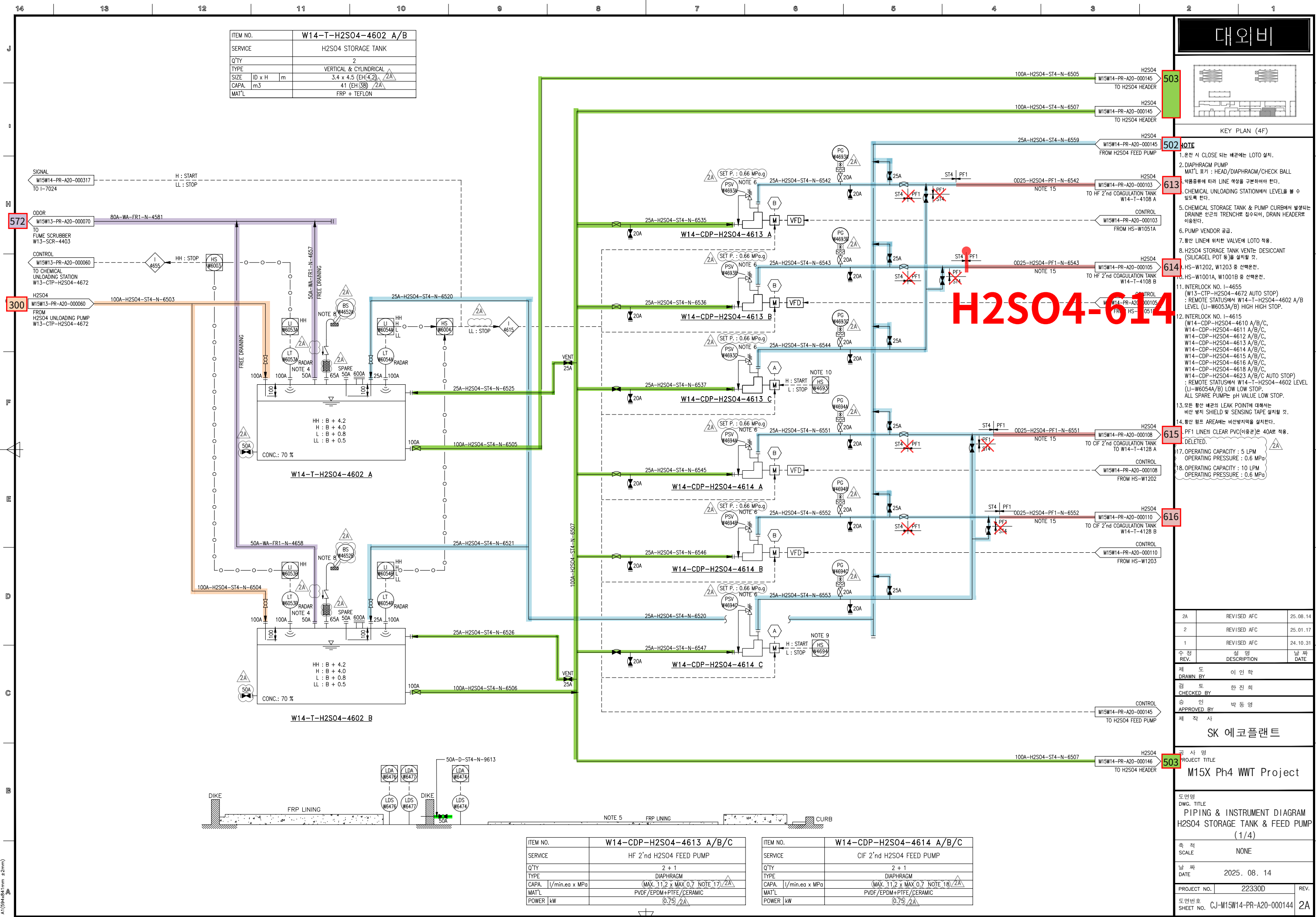


DRAWING LIST FOR TEST PACKAGE PROJECT NO.: 22328D PROJECT TITLE : M15X Ph2 Project		H2SO4-614
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PROJECT NO.: 22328D**PROJECT TITLE : M15X Ph2 Project**

H2SO4-614

[illegible]



ITEM NO.	W14-T-H2SO4-4602 A/B
SERVICE	H2SO4 STORAGE TANK
Q'TY	2
TYPE	VERTICAL & CYLINDRICAL
SIZE	ID x H (EH) 4.2
CAPA.	m3 41 (EH) 39
MAT'L	FRP + TEFLON

ITEM NO.	W14-CDP-H2SO4-4613 A/B/C
SERVICE	HF 2nd H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	l/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 17)
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	0.75

ITEM NO.	W14-CDP-H2SO4-4614 A/B/C
SERVICE	CIF 2nd H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	l/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 18)
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	0.75

대외비

KEY PLAN (4F)

NOTE

1. 온전 시 CLOSE 되는 배관에는 LOTO 설치.

2. DIAPHRAGM PUMP
MAT'L 표기 : HEAD/DIAPHRAGM/CHECK BALL
작동종류에 따라 LINE 색상을 구분하여야 한다.

3. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.

5. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAINE 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이동한다.

6. PUMP VENDOR 공급.

7. 환산 LINE에 위치한 VALVE에 LOTO 적용.

8. H2SO4 STORAGE TANK VENT는 DESICCANT (SILICAGEL POT 등)을 설치할 것.

9. HS-W1202, W1203 중 선택은전.

10. HS-W1001A, W1001B 중 선택은전.

11. INTERLOCK NO. I-4655
(W13-CTP-H2SO4-4672 AUTO STOP)
: REMOTE STATUS에서 W14-T-H2SO4-4602 A/B LEVEL (LI-W6053A/B) HIGH HIGH STOP.

12. INTERLOCK NO. I-4615
(W14-CDP-H2SO4-4610 A/B/C,
W14-CDP-H2SO4-4611 A/B/C,
W14-CDP-H2SO4-4612 A/B/C,
W14-CDP-H2SO4-4613 A/B/C,
W14-CDP-H2SO4-4614 A/B/C,
W14-CDP-H2SO4-4615 A/B/C,
W14-CDP-H2SO4-4616 A/B/C,
W14-CDP-H2SO4-4618 A/B/C,
W14-CDP-H2SO4-4623 A/B/C AUTO STOP)
: REMOTE STATUS에서 W14-T-H2SO4-4602 LEVEL
(LI-W6054A/B) LOW LOW STOP.
ALL SPARE PUMPS는 pH VALUE LOW STOP.

13. 모든 환산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.

14. 환산 펌프 AREA에는 비산방지막을 설치한다.

15. PF1 LINE의 CLEAR PVC(이용관)은 40A로 적용.
DELETED.

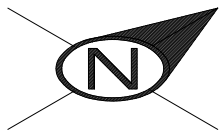
17. OPERATING CAPACITY : 5 LPM
OPERATING PRESSURE : 0.6 MPa

18. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.6 MPa

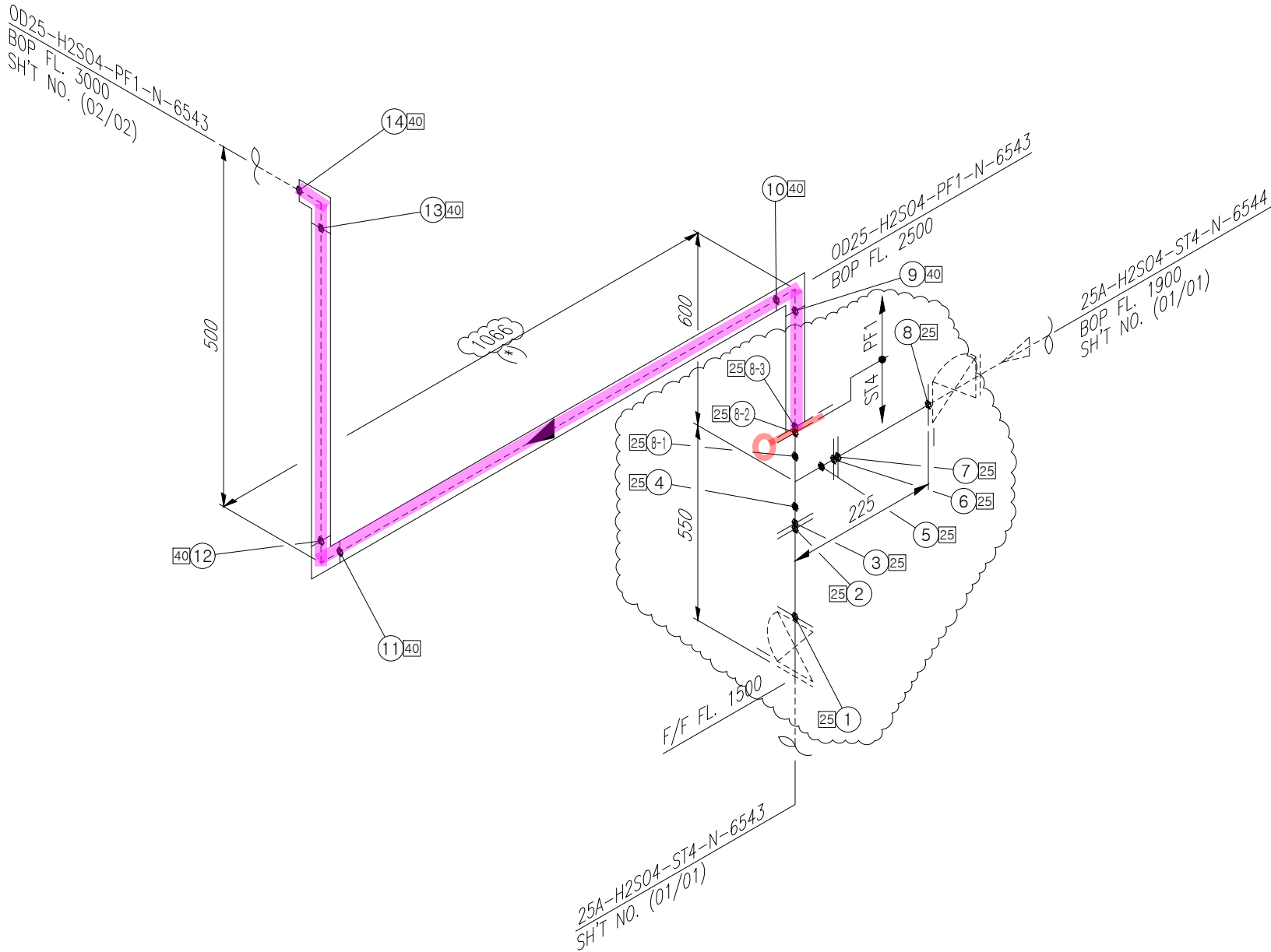
2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정 REV.	실명 DESCRIPTION	날짜 DATE

제 DRAWN BY	이인학
검 CHECKED BY	한진희
승 APPROVED BY	박동영
제 작 사	SK 에코플랜트

제 사 명 PROJECT TITLE	M15X Ph4 WWT Project
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM H2SO4 STORAGE TANK & FEED PUMP (1/4)
속 척 SCALE	NONE
날 짜 DATE	2025. 08. 14
PROJECT NO.	22330D
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000144
REV.	2A



2



- <NOTE>
- 1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
 - 2. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
PF1			
P	OD25	2.2M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	2.2M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)

ST4			
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW - SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW - SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

CLIENT



1

CONTRACTOR		PROJECT NAME	
SK ecoplant		M15X Ph4 Project	
COMMODITY		JOB NO.	
H2SO4		22330D	
LINE NO.		SH'T NO.	REV.
OD25-H2SO4-PF1-N-6543		01 / 02	0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	△	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	6.7	60	UNIT	4FA								
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A									
						REF. P & ID CJ-M15W14-PR-A20-000145		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	COMMODITY	JOB NO.
														H2SO4	22330D
														LINE NO.	SH'T NO.
														OD25-H2SO4-PF1-N-6543	01 / 02
															REV.
															0



COMMODITY	H2SO4	JOB NO. 22330D	
LINE NO.	0D25-H2SO4-PF1-N-6543	SH'T NO. 01 / 01	REV. 0



* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA W14							CONTRACTOR 	PROJECT NAME	
	THICK.	-	DESIGN	6.7	60	UNIT 5FA								M15X Ph4 Project	
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050008-A							COMMODITY H2SO4	JOB NO.	
						CJ-M15W14-PR-A20-000144		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO. 0D25-H2SO4-PF1-N-6543	SH'T NO. 01 / 01	REV. 0
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			

